

Zeinab Dast Mozd (2323135)

CF9697-SU-CO

Big-Data for Computational Finance

Academic Year: 2023/24

Assignment 2: Report

Table of Contents

<i>Introduction</i>	2
<i>Task 1: Linear Regression Model</i>	2
Mean Squared Error for the training data.....	3
Mean Squared Error for the testing data	3
Discussion	3
<i>Task 2: Ridge Regression Model</i>	3
Mean Squared Error for the best model for the train data	3
Mean Squared Error for the best model for the test data	3
Discussion	4
<i>Task 3: Lasso Regression Model</i>	4
Mean Squared Error for the best model for the train data	4
Mean Squared Error for the best model for the test data	4
Discussion	4
<i>Task 4: Random Forest</i>	5
Mean Squared Error for the best model for the train data	5
Mean Squared Error for the best model for the test data	5
the importance of the variables in predicting default.....	5
Discussion	5
<i>Task 5: Neural Network model</i>	5
the accuracy of model for the train data	5
the accuracy of model for the test data	5
why choosing the Neural Network model	5
Discussion	5
<i>References</i>	6

Introduction

Machine learning models

In the field of financial risk management, machine learning models play a crucial role in predicting loan default based on provided features. These models offer a comprehensive approach to assessing the likelihood of loan default, particularly by leveraging various algorithms such as random forest, logistic regression, decision trees, XGBoost, and LightGBM [1]. The predictive accuracy of these models is highly dependent on the demographics of the individuals, with lower income individuals being more likely to default on loans [1]. Additionally, the interpretability of these models varies, with some being more interpretable than others, such as random forest being more interpretable than neural network models but less interpretable than decision trees [1]. The use of machine learning in this context is essential for effective risk management in the financial sector, providing insights into the prediction of loan default based on historical data and key features [2]. In predicting loan default, understanding the correlation between variables and loan status is crucial. The strength and direction of the relationship between two variables are quantified using the linear correlation coefficient, also known as Pearson's product moment correlation coefficient. This statistic numerically describes the strength and direction of the linear relationship between two variables, always ranging between -1 and +1. The correlation coefficient provides insights into how strongly the variables are related and the direction of their relationship [3]

Most and Least Correlated Variables

The most correlated variables with loan status are (ApplicantIncome, LoanAmount) and (Credit_History, Loan_Status). Additionally, LoanAmount is correlated with CoapplicantIncome. On the other hand, the least correlated variables with loan status are not explicitly mentioned in the provided search results. However, further analysis of the correlation between independent variables and loan status can provide valuable insights for predicting loan default [4]. In the context of predicting loan default, understanding the correlation between various features and the loan status is essential for building accurate machine learning models. By identifying the most and least correlated variables, it becomes possible to prioritize the most influential features for predicting loan default and to refine the predictive models accordingly.

Task 1: Linear Regression Model

Linear regression models offer a transparent and interpretable framework for understanding the relationship between borrower attributes, loan characteristics, and the probability of default. By leveraging linear regression, financial institutions can gain valuable insights into the factors influencing loan default and enhance their risk assessment strategies, ultimately enabling informed decision-making regarding loan approvals and risk mitigation strategies. In this assignment, I used three methods to manage missing values:

- **The first method:**

For numerical columns, I filled the missing values with the median. Since, It's robust to outliers and skewed distributions, common in financial data (dti, revol_util, etc.) For the 'emp_length' column, I filled the missing values with the mode. As, it Preserves the most common value, maintaining the central tendency of the data. And I dropped two columns, 'member_id' and 'id', because they did not provide any useful information and all their values were missing. By applying this method, I was able to effectively handle the missing data and improve the dataset's quality for further analysis. Then I applied linear regression model, and I got this result.

Mean Squared Error for the training data = 0.06780941597891035
Mean Squared Error for the testing data = 1.693199897110197e+18

- **The second method:**

I dropped three columns: ['id', 'member_id', 'mths_since_last_delinq'] because the first two Uninformative and 'last_delinq' have enormous number of missing values approximately half of its value and I removed the rows that contained missing values. After Applying the model, I got this result.

Mean Squared Error for the training data = 0.0654716056696797
Mean Squared Error for the testing data = 6.215507141425878e+16

- **The third method:**

In the third method, I remove the 'id' and 'member_id' and filled 'mths_since_last_delinq' with 0 values and then dropped all the rows with missing values. I got this result.

Mean Squared Error for the training data = 0.06546840324699656
Mean Squared Error for the testing data = 2.061948554476924e+18

Discussion

In the first method of filling missing value the result was acceptable on the training data, but it was not good on the test data. This means the model could not make good predictions or generalize well on unseen data. After employing this method of filling values, I observed that the results were better compared to the second and third approaches. However, I will not continue with this method because I believe this data is different from weather data. Each client has unique features and corresponding labels, and filling missing values with the mean or other methods could distort the data. Therefore, it is best to either delete the missing values or fill them with zero. This is the approach I will pursue moving forward. Then, used another method of dealing with missing values and got different results. There was a noticeable change in the mean squared error on the test data, which means this method of handling missing values does not improve the model's performance. The third approach resulted in better performance compared to method 2, as indicated by the improved mean squared error on the test data. I will now attempt to apply all models using methods 2 and 3.

Task 2: Ridge Regression Model

Ridge Regression is a type of linear regression that includes a regularization term to prevent overfitting. λ controls the strength of the regularization. When $\lambda=0$, Ridge Regression becomes standard linear regression. As λ increases, the impact of the penalty term grows, leading to smaller coefficient values. It helps to manage multicollinearity (when independent variables are highly correlated). It reduces model complexity by shrinking the coefficients, which can lead to better generalization on unseen data. I used the second method and the third method, and I got this result:

Second method

Best Alpha: 0.01
Mean Squared Error (Training data): 0.06547170890739468
Mean Squared Error (Testing data): 0.06620607027686866

Third method

Mean Squared Error (Training data) for Ridge Regression: 0.06546797170470291

Mean Squared Error (Testing data) for Ridge Regression: 0.06620194985396584

Best alpha (lambda) value: 0.01

Discussion

The result was approximately the same, but there was a huge improvement in the mean squared error on the test data compared to the previous model in both methods were used to fill the missing values. And that because of that Ridge Regression modifies the linear regression model by adding a penalty term to the loss function, which is the sum of the squares of the coefficients. This regularization technique helps to prevent overfitting, which was happening in the previous model, especially in cases where the features are highly correlated. The regularization parameter λ controls the strength of the penalty, balancing the trade-off between fitting the training data well and keeping the model coefficients small to ensure better generalization to new data.

Task 3: Lasso Regression Model

Lasso Regression modifies the linear regression model by adding a penalty term based on the absolute values of the coefficients. This regularization technique not only helps prevent overfitting but also enables automatic feature selection by driving some coefficients to zero. The regularization parameter λ governs the strength of this penalty, allowing for a balance between model complexity and predictive performance.

In the second method the best model was

Best Alpha: 0.01

Mean Squared Error (Training data): 0.06762740839251773

Mean Squared Error (Testing data): 0.068632463967141

In the third method was

Best alpha (lambda) value: 0.01

Mean Squared Error (Training data): 0.06762740839251773

Mean Squared Error (Testing data): 0.06840822181278264

Discussion

In each method was used to handle missing values, Both MSE values are quite close to each other. This suggests that the model is generalizing well and no overfitting. The best lambda value is **0.01**. This parameter controls the strength of the regularization. A small value like 0.01 indicates that the model is balancing well between fitting the data and applying regularization to prevent overfitting. Overall, the results show that the Lasso Regression model performs well, with approximately similar MSE values for both training and testing data. The Ridge model often outperforms the Lasso model (Lasso regression) when dealing with datasets that have many features, especially when those features are correlated. In contrast, Lasso regression tends to reduce the number of features by setting some coefficients to zero, which enhances model performance when features are not highly correlated. Therefore, the superior performance of Ridge regression in this context indicates that the features in our dataset are indeed highly correlated.

Task 4: Random Forest

Random Forest is an ensemble learning method primarily used for classification and regression tasks. In predicting loan defaults Random Forest might consider features like credit score, income level, loan amount, and employment status. By aggregating multiple decision trees, the model can accurately predict the likelihood of defaulting while also indicating which features are most influential in the decision-making process.

In the second method the best model was

Best n_estimators: 200

Mean Squared Error (Training data): 0.002713125795087582

Mean Squared Error (Testing data): 0.8867148883690429

In the third method was

Mean Squared Error (Training data) for Lasso Regression: 0.0027307095850866037

Mean Squared Error (Testing data) for Lasso Regression: 0.8867148883690429

Best n_estimators: 200

Feature importance These values represent how much each feature contributes to the model's predictions. A higher value indicates that the feature is more important in predicting the target variable. Like feature 43 and 46 which have high score and there are features have 0 score that mean they do not have affect. I used another method of missing value by keeping all the columns and fill missing values with 0 and I notice that the features correspond to the index because id and member id correspond to feature0 and feature1 and they both have 0 importance from that observation I can say that feature43 corresponds to emp_length_9 years. And 26 to home_ownership_NONE. And there is overfitting, and the model does not predict well however I tried to use normalizer instead of scaler for normalization and the performance was 0.03 and it was the best, better than neural network model. I tried to use 300 and more but it took too much time even one hour, the result was the same just less overfitting.

Task 5: Neural Network model

I chose MLPRegressor model for predicting loan defaults. this task is a classification problem rather than a regression one but this model suitable for both. Predicting who will default on a loan involves determining whether a borrower will default (yes or no), which is a classification task. After trying this model, I got high accuracy and that is why I chose it and there are other reasons like its activation functions introduce non-linearity, enabling the network to learn complex patterns and the model learns by adjusting weights to minimize the loss function through backpropagation and gradient descent.

Second method *Accuracy (Training data) for NN: 0.970197671004990*

Accuracy (Testing data) for NN: 0.9642888603952352

Third method *Accuracy (Training data) for Neural Network: 0.9707016342107838*

Accuracy (Testing data) for Neural Network: 0.9627227969892431

Discussion

In both method of filling missing values, the accuracy was high, and the mean squared error was less. therefore, it the best model on predicting. It is better than all the previous models and for all models the third method of filling missing values was better. And for the most 10 correlated variables with the loan status and less 10 I will put the following

10 Most Correlated Variables:	10 Least Correlated Variables:
1. grade_E: Loan grade E indicating a higher risk category with a specific correlation to default. 2. grade_F: Loan grade F indicating an even higher risk category with a specific correlation to default. 3. grade_D: Loan grade D indicating a moderate-high risk category with a specific correlation to default. 4. grade_B: Loan grade B indicating a low-moderate risk category with a specific correlation to default. 5. grade_G: Loan grade G indicating the highest risk category with a specific correlation to default. 6. application_type_Joint App: Loan application made by two borrowers together. 7. home_ownership_RENT: Borrower is renting their home. 8. home_ownership_MORTGAGE: Borrower has a mortgage on their home. 9. grade_C: Loan grade C indicating a moderate risk category with a specific correlation to default. 10. emp_length_10+ years: Borrower has been employed for over 10 years.	1. tot_coll_amt: Total amount of collections (debt recovery) on the borrower's account. 2. loan_amnt: The amount of money borrowed in the loan. 3. annual_inc: The borrower's yearly income. 4. total_rev_hi_lim: The total credit limit available to the borrower. 5. emp_length_5 years: The borrower has been employed for 5 years. 6. installment: The regular payment amount the borrower must make to repay the loan. 7. emp_length_< 1 year: The borrower has been employed for less than 1 year. 8. total_rec_prncp: The total principal amount repaid by the borrower. 9. id: A unique identifier for the loan record. 10. member_id: A unique identifier for the borrower.

References

- [1] Reddy, I., Madhavi Nirati and K. Venkatesh Sharma (2022). Predicting Possible Loan Default Using Machine Learning. *International journal of computer engineering in research trends*, 9(12), pp.244–252.
- [2] Maddali, S. (2023). *suhasmaddali/Predicting-Loan-Default-Using-Machine-Learning*. [online] GitHub. Available at: <https://github.com/suhasmaddali/Predicting-Loan-Default-Using-Machine-Learning>.
- [3] Kiernan, D. (2014). Chapter 7: Correlation and Simple Linear Regression.
- [4] Analytics Vidhya. (2022). *Loan Prediction Problem from Scr*

